

CPP NOTES – DAY 25

OOPS

Concepts:

1. Objects
2. Classes
3. Abstraction
4. Encapsulation
5. Inheritance
6. Polymorphism
7. Exception handling

Classes

A class is a user-defined data type, which holds its own data members and member functions that can be accessed and used by creating an instance of that class. A C++ class is like a blueprint for an object.

For Example: Consider the Class of Cars. There may be many cars with different names and brands but all of them will share some common properties like all of them will have 4 wheels, Speed Limit, Mileage range, etc. The car can also accelerate, turn, apply brakes, etc. So here, the Car is the class, wheels, speed limits, and mileage are its attributes (data members) and accelerate, turn, apply brakes are its methods (member functions).

```
class GfG {  
public:  
  
    // Data member  
    int val;  
  
    // Member function  
    void show() {  
        cout << "Value: " << val << endl;  
    }  
};
```

FYI:

- *Exception handling is used to end the program in a proper way without causing any bugs or error which can cause a crash.*
- *Protected and private(access modifiers) are part of inheritance.*